

Local Network-Based Influence Prediction

Study Document — Pipeline, Results, and Glossary

Working reference for the project team

0. How to use this document

This is the working reference for the project. It explains **what we are doing, why each step exists, what tool does each job, and what we have measured so far.**

Part VIII is a complete glossary. Every technical term and every mathematical expression used anywhere in this document is defined there in one or two plain sentences. If a word or symbol is unfamiliar, look it up there first — nothing is assumed.

Three conventions used throughout:

- **Inputs** are always *local* — computed from a bounded neighbourhood around a node.
- **Targets** are always *global* — computed from the whole network, and are what we try to predict.
- **Baselines** are rival methods we must beat, or honestly report losing to.

The single rule that protects the entire project: **a global measurement may never appear as an input feature.** Breaking it invalidates every result downstream. This rule is enforced in code (`assert_no_leakage`), not just by discipline.

Nothing in this project is inherited. We take no published result, no reported number, and no claim about which radius works as given. Algorithm *definitions* are specifications we implement ourselves and then verify against an independent reference. Every number in Part VII is one we measured.

PART I — THE PROBLEM

1. The research question

Can we predict how influential a person is in an **entire** network, using only information visible in their **local neighbourhood** — their friends, friends-of-friends, and perhaps one hop further?

1.1 Why the constraint matters

Real networks are too large to process completely. An entire social platform, a national telecom graph, or the web cannot be loaded into memory and analysed globally. Yet influence is defined globally. If a bounded local view can recover global importance, the method scales to networks where the global computation is impossible.

1.2 Our data: real network corpora

We train and test on **real networks already used in this literature**, rather than generating our own. The reason is comparability: this subfield shares a benchmark convention — spreading-process ground truth, Kendall τ and top-k overlap, roughly ten to twelve real networks. Adopting that convention exactly makes our numbers directly comparable to published ones instead of only internally consistent.

Networks used so far (both from the SNAP pilot set, both processed by our own pipeline):

Network	Nodes (after cleaning)	Edges	Mean degree	Character
ca-GrQc	4,158	13,422	6.46	Sparse collaboration network
email-Eu-core	986	16,064	32.58	Dense communication network

These two were chosen deliberately as a contrast: one sparse, one dense. As Part VII shows, that contrast turned out to matter a great deal.

Corpus to expand into: the remaining SNAP pilot set, the Erkol network list, the Banerjee village networks, Netzschleuder, and ICON. Synthetic generation remains available in the plan (Part V) but is deferred — it buys structural control, whereas real corpora buy comparability, and comparability comes first.

Scale rationale. We work at a scale where the true global influence remains exactly computable. That computability is essential: it is the ground truth our local predictions are graded against.

1.3 The core vocabulary

Full definitions are in Part VIII. The four terms you cannot read further without:

Term	Meaning
Hop	One step along an edge. 1-hop = direct friends; 2-hop = friends-of-friends.
r-ball	Everything within r hops of a node.
Ego-network	A node (the <i>ego</i>), its direct friends (<i>alters</i>), and all edges among them.
Local feature	A number computable from a bounded neighbourhood only.

2. Defining influence: three targets

"Influence" is vague, so we pin it to three measurable quantities. A node can score high on one and low on another — that disagreement is itself a finding.

2.1 Target A — Bridge influence (betweenness centrality)

$$b(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$$

Plain meaning: the fraction of everyone's shortest paths that route through you. High betweenness means you are a broker between otherwise-separate groups; remove you and the network fragments.

Why it is global: to know which shortest paths pass through v , you must know shortest paths between *every pair* in the network. No local patch contains this.

Why it is not just popularity: a node with two friends can have high betweenness if those two friends belong to different regions of the network.

2.2 Target B — Spread influence (expected cascade size)

$$\sigma(i) = E[|R(i)|]$$

Plain meaning: start a rumour at node i , let it spread at random, and count how many people eventually hear it. Average that over many runs.

Because it is stochastic, we estimate by **Monte Carlo**:

$$\hat{\sigma}(i) = (1/M) \sum_{m=1..M} |R_m(i)| \quad SE = s / \sqrt{M}$$

M is the number of runs and s the sample standard deviation. We choose M so that each node's standard error is far below the spread *between* nodes — otherwise we would be ranking simulation noise rather than real differences. Our runs report a ratio of 0.0036, comfortably safe.

2.3 Target C — Spread volatility

$$CV(i) = sd(|R(i)|) / \sigma(i)$$

Plain meaning: two seeds can have the same average reach but behave completely differently — one reaches ~200 people almost every time, another usually reaches 5 but occasionally 800. CV measures that unreliability on a scale-free basis.

Why it matters: the averaged number erases the difference between a *reliable broadcaster* and a *lottery ticket*, which is exactly the distinction a practitioner cares about.

The trap, and the control. Variance is mechanically coupled to the mean — a seed reaching 500 has more room to vary than one reaching 3. Predicting raw variance would largely re-predict the mean in disguise. Two controls, both implemented:

1. Predict **CV** rather than raw variance (scale-free).

2. Predict the **variance residual**: fit the expected spread-versus-mean relationship non-parametrically, then predict only what is left over.

$$\text{Var}(i) = f(\sigma(i)) + \varepsilon_i \rightarrow \text{predict } \varepsilon_i \text{ from local features}$$

2.4 The Independent Cascade mechanism

The spreading rule we simulate:

1. The seed node starts active; all others inactive.
2. When a node *first* becomes active, it gets **one** attempt to activate each inactive neighbour, succeeding independently with probability **p**.
3. Success or failure, that attempt is never repeated.
4. Newly activated nodes act in the next round.
5. Repeat until a round produces no new activations. Count the total activated.

Edge probability conventions. The convention silently changes results, so it is an explicit parameter, never a buried default:

Convention	Rule	Note
uniform	p on every edge	Simplest; inflates hub influence
weighted cascade	$p_{uv} = 1/\text{degree}(v)$	Normalises each node's incoming pressure
trivalency	p drawn per edge from	Standard in benchmarking

We report under at least two. A spreading process with a rate parameter is a fourth convention that does not map exactly onto any of them.

3. Choosing p: the epidemic threshold

This was a real problem and it has a principled solution.

Why the obvious approaches fail. Fixing $p = 0.05$ everywhere is meaningless: on a dense network that is far above the tipping point (everything reaches everyone) and on a sparse one far below (nothing spreads). Comparing networks at fixed p compares different dynamical regimes. Alternatively, picking the p that maximises the spread of influence scores is confounded with scale — larger p means larger cascades means larger raw variance, so the criterion just chases the top of whatever range you searched. **We confirmed this failure empirically before discarding the method.**

The solution. Compute each network's own critical threshold and work at a fixed *multiple* of it. Then " $p = 1.5 \times \beta_c$ " means the same dynamical regime on every network, and cross-network comparison is finally valid.

3.1 The non-backtracking operator

The naive estimate $\beta_c = 1/\lambda_1(A)$ using the adjacency matrix is poor on networks with hubs, because the leading eigenvector localises on a hub and drags the estimate around. The better operator tracks walks that never immediately step back where they came from:

$$B[(i \rightarrow j), (k \rightarrow l)] = 1 \text{ if } j = k \text{ and } i \neq l, \text{ else } 0$$

$$\beta_c = 1 / \lambda_1(B)$$

B is $2m \times 2m$, which is large. The **Ihara-Bass identity** gives the same leading eigenvalue from a $2n \times 2n$ matrix:

$$M = \begin{bmatrix} A & I - D \\ I & 0 \end{bmatrix}$$

where A is adjacency, D the degree diagonal, and I the identity. We implemented both, and verified they agree to $1e-9$.

3.2 What we measured

Network	β_c (non-backtracking)	β_c (adjacency)	β_c (mean-field)
ca-GrQc	0.02250	0.02192	0.05889
email-Eu-core	0.01338	0.01311	0.01358

Finding. On the dense network all three agree. On the sparse network the mean-field estimate is wrong by a factor of 2.6. This is concrete evidence that the non-backtracking operator is a necessity rather than a refinement — and it is our own measurement, not an inherited claim.

Finding. Sweeping p as a multiple of each network's own β_c , the between-node coefficient of variation *peaks near criticality* (roughly $1.0\text{--}1.5 \times \beta_c$) and falls away on both sides. Discriminability between nodes is highest near the tipping point, which is where we therefore run.

PART II — THE FEATURES

4. What makes a feature "local"

A feature is local if computing it for node v requires only the r -ball around v , for small r . Degree is local (count your own edges). Betweenness is not (requires all-pairs shortest paths).

Two further quality criteria: a feature must **vary across nodes** (a constant teaches nothing), and must not be a **trivial duplicate** of another.

5. The two tagging axes

Every feature carries two tags, and this is what makes the experiments cheap.

Hop tag — the smallest radius that can compute it:

- hop 0: the node's own edges only
- hop 1: the node + its neighbours + edges among them (the ego network)
- hop 2: adds friends-of-friends
- hop 3: adds one ring further

Tier tag — what object the feature describes:

- `node` : properties of the node itself
- `edge` : properties of its incident edges, aggregated
- `subgraph` : counts of small structures it participates in

Radius and tier are independent axes. That is deliberate: it lets us ask whether it is better to look **further** (more hops) or look **richer** (more tiers) for the same compute budget.

Because of the tagging, the radius sweep is a one-line filter (`select_features`) rather than a rewrite.

6. The implemented feature set (43 features)

6.1 Hop 0 — the node itself

Feature	Meaning
<code>degree</code>	Number of direct neighbours
<code>degree_normalized</code>	$\text{degree} / (n-1)$; a rescaling for cross-network comparability

6.2 Hop 1 — ego-network structure

Feature	Meaning
<code>clustering_coefficient</code>	Fraction of your neighbour pairs that are themselves connected
<code>triangle_count</code>	Triangles through the node
<code>ego_net_size</code>	Nodes in the ego network (degree + 1)
<code>ego_net_edges</code>	Edges inside the ego network
<code>ego_net_density</code>	Ego-net edges / possible edges

Feature	Meaning
edges_leaving_ego	Edges from your neighbours to nodes outside your ego net
effective_size	Burt's measure: count of non-redundant contacts
boundary_porosity	Outward edges / all ego-net edge endpoints — how "leaky" your boundary is

`boundary_porosity` is included deliberately: a node cannot see past its own boundary, but it *can* observe how porous that boundary is. This is the key signal for the self-assessment question in Angle 5.

6.3 Hop 1 — neighbour aggregates

Each of `neighbour degree`, `neighbour clustering`, `neighbour triangles` summarised by mean / max / min / std / sum, giving fifteen features. Plus:

Feature	Meaning
assortativity_local	Mean neighbour degree minus own degree — do you connect "up" or "down"?

6.4 Hop 1 — brokerage

Feature	Meaning
ego_betweenness	Betweenness of the ego computed <i>inside its own ego network only</i>

Our derivation. Within an ego network, two neighbours j, k of the ego are either directly connected (shortest path length 1, cannot pass through the ego) or not (every shortest path has length 2 and runs through a common neighbour; the ego is always one such). So for each non-adjacent pair, the contribution is $1/(\text{number of their common neighbours inside the ego net, plus the ego itself})$. Summing gives exact ego-betweenness without running full betweenness on each ego network. Verified against brute force to $2e-11$.

6.5 The H-index ladder (hop-parameterised)

$$h^{(0)}(i) = \text{degree}(i)$$

$$h^{(n+1)}(i) = H(\{ h^{(n)}(j) : j \text{ a neighbour of } i \})$$

where **H** is the H-operator: the largest h such that at least h of the values are $\geq h$ — the same operator as a researcher's h -index, applied to neighbours instead of citations.

This is a radius ladder in disguise: $h^{(n)}$ depends on information exactly n hops away, so $h^{(1)}$ is a hop-1 feature, $h^{(2)}$ a hop-2 feature, and so on.

We tested rather than assumed the convergence claim. Iterating the operator moves toward the k -core number: agreement rose from 49.4% at order 1 to 93.9% at order 12 on email-Eu-core. That is our measurement.

6.6 Collective Influence (radius-parameterised)

$$CI_{\ell}(i) = (k_i - 1) \cdot \sum_{j \text{ at exactly distance } \ell \text{ from } i} (k_j - 1)$$

Note "exactly distance ℓ " — the shell, not the ball. A genuinely local score with the radius as an explicit parameter, which makes it the natural classical competitor in the radius sweep.

6.7 Multi-hop shells (hops 2-3)

Feature	Meaning
shell_r_count	Nodes at exactly r hops
shell_r_degree_mean / _max	Popularity at exactly r hops
growth_ratio_r	Shell r size / shell r-1 size — how fast your reachable set expands
reach_within_r	Total distinct nodes within r hops

6.8 What is NOT a feature

Quantity	Role
betweenness	Target (bridge influence)
spread_mean , spread_cv , spread_resid	Targets (spreading influence)
closeness , pagerank , eigenvector , katz , harmonic , coreness	Rival baselines only — never inputs

Why these are global. Betweenness and closeness depend on shortest paths across the whole network. PageRank and Katz are recursive — your score depends on your neighbours' scores, which depend on theirs, unravelling across the entire graph. None can be computed from a bounded local patch.

PART III — EVALUATION

7. Metrics

Influence is heavily skewed: a few enormous nodes, thousands near zero. Raw error is dominated by the large ones and hides whether the *ranking* is right, which is what actually matters for finding influencers.

7.1 Primary

Kendall's τ

$$\tau = (n_c - n_d) / (\frac{1}{2} n(n-1))$$

Over all pairs of nodes, n_c is the count ranked in the same order as truth (concordant) and n_d the count reversed (discordant). $\tau = +1$ is a perfect ranking, 0 is random, -1 is exactly reversed.

Precision@k

$$P@k = |top-k \text{ predicted} \cap top-k \text{ true}| / k$$

Of the truly top-k influencers, how many did we catch? This is what a practitioner actually asks.

7.2 Secondary

Spearman correlation; RMSE reported but never headlined.

7.3 Honest error estimation

k-fold cross-validation with out-of-fold prediction. Nodes are split into k groups; we train on k-1 and predict the held-out group, rotating until every node has a prediction from a model that never trained on it. Without this, a flexible model partly memorises its training nodes and its errors look artificially small — which would wreck the failure analysis, since we would be studying memorisation artefacts rather than genuine structural blind spots.

Known limitation, stated openly. On a *single* graph, held-out nodes' features are still computed on a graph containing the training nodes. This is not fully independent transfer. The clean version is training on some networks and testing on entirely separate ones; the pipeline supports this and it is the next step as the corpus grows.

8. The baseline ladder

Every result is reported against this ladder:

1. **Degree only** — the naive "popularity = influence" rule. *This is the $r=0$ row of every sweep.*
2. **Classical local scores** — H-index ladder, Cl_l , ego-betweenness.
3. **Our full feature set** — node + edge + subgraph tiers across hops.
4. **Published baselines** — *reimplemented by us, or run from released code.*

On rung 4 — the replication rule. Do not accept a published number as a baseline until you can reproduce it. Authors may cherry-pick; results may not replicate. Check for released code first; if unavailable, implement it yourself. Replication is itself respected work.

PART IV — THE IMPLEMENTED PIPELINE

9. Module structure

The codebase is a Python package, `influence/`, of five modules.

Module	Responsibility
<code>preprocessing.py</code>	Loading and the cleaning protocol; the <code>Network</code> container
<code>features.py</code>	All 43 local features, with hop/tier tagging and per-group timing
<code>dynamics.py</code>	IC and SIR simulators, edge-probability conventions, full distributions
<code>criticality.py</code>	Non-backtracking operator, Ihara-Bass form, β_c
<code>targets.py</code>	Ground truth, variance residual, and the leakage guard
<code>experiment.py</code>	Metrics, out-of-fold prediction, the radius $\times$ richness sweep

Runner scripts: `stage1_prepare.py` (checkpoints the expensive deterministic work), `stage2_sweep.py` (runs the grid, saving each cell as it completes so an interruption never loses finished work), `analyse.py`, `make_fig1.py`.

10. The cleaning protocol

Six decisions, applied identically to every network and logged into a provenance record that goes straight into the methods section:

1. **Directedness** $\rightarrow$ symmetrize. Stated, not silent.
2. **Largest connected component** $\rightarrow$ restrict to it. Influence on a small fragment is capped by fragment size, which would contaminate the ground-truth distribution.
3. **Self-loops** $\rightarrow$ removed; they make the non-backtracking construction ill-defined.
4. **Multi-edges** $\rightarrow$ collapsed to simple edges.
5. **Weights** $\rightarrow$ discarded entirely. Half-using weights is the most common silent error in this literature.
6. **Node IDs** $\rightarrow$ reindexed to $0..n-1$, with the mapping **kept**, so results can always be joined back to the source file's labels.

11. The percolation shortcut

The single most important performance decision in the pipeline.

The problem. Naively, M simulations from each of n seeds is $O(n \cdot M)$ cascades — millions of runs.

The insight. Independent Cascade with a fixed per-edge probability is exactly **bond percolation**. Each edge either transmits or does not, independently, and that draw does not depend on who is seeding. So we can sample one live-edge subgraph, find its connected components, and every node's reached set for that sample *is* its component.

One run therefore yields a sample for **all n seeds simultaneously** — M percolation runs instead of $n \cdot M$ cascades, a speedup of roughly n .

Measured: 1.6s for all 986 seeds, versus 36s for just 25 seeds by direct simulation — about 900 $\times$.

Scope. Exact for symmetric conventions (uniform, trivalency). It does *not* hold for weighted cascade, where the probability depends on the receiving node's degree and the edge is asymmetric. Both paths are implemented, and we verified they agree where both are valid.

12. Verification

Nothing is assumed. Each implementation was checked against an independent reference:

Implementation	Checked against	Result
degree, clustering, triangles	networkx	exact match
H-operator	hand-computed unit tests	correct
ego-betweenness	brute-force betweenness on explicit ego subgraphs	2e-11
lhara-Bass eigenvalue	direct $2m \times 2m$ non-backtracking matrix	1e-9, three graph families
Brandes betweenness	networkx	1e-14
igraph fast betweenness	our own Brandes	1e-14
percolation shortcut	direct per-seed IC simulation	$r = 0.9993$
SIR with $\mu=1$	Independent Cascade (they should coincide)	0.4% difference
leakage guard	deliberate injection of forbidden names	all caught, no false positives

On the igraph fast path. We wrote Brandes' algorithm ourselves first and confirmed it matches. Using the C implementation afterwards for large graphs is a speed decision, not an inherited result — the algorithm is a specification and we have verified our reading of it.

PART V — THE PHASES

Phase	Status
0 — Reproduce the supervisor's baseline plot	Done — the $P(r)$ curve exists
1 — Cleaning protocol and loader	Done
2 — Three-tier feature engine	Done (43 features)
3 — Ground truth: β_c , cascades, betweenness	Done
4 — Locality budget sweep	Done on two networks
5 — The analysis angles	Angles 1 and 2 partly done; 3-6 pending
6 — Corpus expansion	Next
7 — Neural comparison	Last, per supervision

13. Angles

Angle 1 — Locality budget. How far out must you look? Delivered: $P(r)$ curves, $r^*(\epsilon)$ across tolerances, marginal gain per hop. See Part VII.

Angle 2 — Multi-target contrast. Same features, three targets. Delivered in part — the striking result is that the three targets have *different* horizons.

Angle 3 — Failure atlas. Which nodes does local prediction fail on, and why? Out-of-fold residuals are already computed inside the sweep; the remaining work is retaining and profiling them.

$$\Delta(v) = \text{rank}_{\text{true}}(v) - \text{rank}_{\text{pred}}(v)$$

Large positive Δ = under-predicted "hidden influencer."

Angle 4 — Robustness under noise. Delete a fraction ρ of edges, recompute features on the damaged graph, predict, and compare against the *clean* truth. The comparison that makes it a result: does the local model track truth better than recomputing the global score on the broken graph?

$$\tau_{\text{local}}(\rho) > \tau_{\text{recompute}}(\rho) ?$$

Angle 5 — The locality gap. How much of a node's importance physically lies beyond r hops?

$$g_r(v) = 1 - b_r(v) / b(v) \in [0, 1]$$

where $b_r(v)$ counts only pairs with both endpoints inside the r -ball. Then: can a node predict its *own* gap from local features? It cannot see past its boundary, but `boundary_porosity` and the growth ratios describe how leaky that boundary is. Already implemented as features, so the experiment is ready to run.

Angle 6 — Spread volatility. Partly delivered: `spread_cv` is already a target in the sweep, and its horizon result is in Part VII. Remaining: the variance-residual target and the archetype map.

14. Deferred: synthetic network generation

Not abandoned, deferred. Generated networks (Erdős-Rényi, Barabási-Albert, Watts-Strogatz, Holme-Kim, stochastic block model, LFR) would give structural control and ground truth by construction — in a block model we *know* which nodes are the planted bridges. They would also allow the criticality sweep: measure r^* as p moves through β_c and test whether the required radius peaks at the tipping point.

The trade-off: real corpora buy comparability with published work; generated networks buy control. Comparability comes first.

PART VI — TECHNOLOGY STACK

Tool	Role	Notes
NumPy / SciPy	Arrays, sparse matrices, eigenvalues, connected components	<code>scipy.sparse</code> is the backbone; <code>sparse.linalg.eigs</code> powers β_c
pandas	Feature and result tables	
python-igraph	Fast exact betweenness on large graphs	C-backed; used only after verifying against our own implementation
networkx	Verification reference, graph generators	Pure Python and slower; ideal as an independent check
scikit-learn	Random forest regressor, k-fold CV, metrics	
matplotlib	Figures	
ORCA (planned)	Graphlet orbit counts for richer subgraph features	Single C++ file; sidesteps Python's threading limits
PyTorch Geometric (Phase 7)	GraphSAGE for the neural comparison	Respects locality by construction

15. Performance notes

- **Python threading does not parallelise CPU-bound work** (the GIL). Use processes, or C-backed libraries that release it.
- The development machine had **one CPU**, so `n_jobs=-1` was pure joblib overhead; it is set to 1.
- **Checkpoint everything.** `stage1_prepare.py` saves features, targets, cascades and metadata; `stage2_sweep.py` appends each grid cell as it finishes. A run that dies loses nothing. This was learned the hard way.
- **Cache aggressively.** Features and simulation results are deterministic given a seed.
- **Log wall-clock cost per feature group** during extraction — the cost-versus-quality analysis depends on it and it cannot be reconstructed afterwards.

16. Reproducibility practices

- Fix and record random seeds for every stochastic component.
- Store **full cascade distributions**, not summary statistics — volatility work needs them and re-running is expensive.
- Keep the feature registry in sync with the feature table; it carries the hop and tier tags that the sweep depends on.
- Record the provenance dict for every network.

PART VII — RESULTS SO FAR

17. The locality budget curve

Kendall τ , out-of-fold, full feature richness.

ca-GrQc ($n = 4,158$, $\langle k \rangle = 6.5$, $p = 1.5 \times \beta_c$):

radius	spread (mean)	betweenness	spread volatility
0	0.7809	0.5443	0.4570
1	0.9203	0.9152	0.7179
2	0.9414	0.9162	0.8063
3	0.9428	0.9163	0.8209

email-Eu-core ($n = 986$, $\langle k \rangle = 32.6$, $p = 1.2 \times \beta_c$):

radius	spread (mean)	betweenness	spread volatility
0	0.8456	0.7433	0.7632
1	0.9525	0.9016	0.8963
2	0.9533	0.9013	0.8988
3	0.9535	0.8996	0.8987

18. Finding 1 — different targets have different horizons

Marginal gain in τ from each extra hop, ca-GrQc:

Target	base ($r=0$)	$r0 \rightarrow r1$	$r1 \rightarrow r2$	$r2 \rightarrow r3$
betweenness	0.5443	+0.3709	+0.0010	+0.0002
spread mean	0.7809	+0.1394	+0.0211	+0.0014
spread volatility	0.4570	+0.2609	+0.0884	+0.0146

Betweenness saturates at $r = 1$. Spread mean saturates at $r = 2$. **Volatility is still climbing at $r = 3$.**

This is the headline. Measured on identical features and the same graph, different notions of influence require genuinely different amounts of local information. The published habit of fixing a radius ad hoc — 2 hops here, 3 there — cannot be right for all targets at once.

19. Finding 2 — r^* depends on the tolerance, so report the curve

$r^*(\epsilon)$, smallest radius reaching $(1-\epsilon)$ of the ceiling:

Network	Target	$\epsilon=0.20$	$\epsilon=0.10$	$\epsilon=0.05$	$\epsilon=0.02$	$\epsilon=0.01$
ca-GrQc	betweenness	1	1	1	1	1
ca-GrQc	spread mean	0	1	1	2	2
ca-GrQc	spread volatility	1	2	2	2	3
email-Eu-core	all three	0-1	1	1	1	1

A single r^* is an artefact of an arbitrary tolerance — exactly the ad hoc choice this project criticises. **We report the full $P(r)$ curve and the marginal-gain table, with $r^*(\epsilon)$ as a summary across a range of ϵ .**

20. Finding 3 — depth beats richness decisively

At every radius on both networks, adding edge and subgraph tiers changes τ by less than ± 0.001 , sometimes negatively. One extra hop buys up to +0.37.

ca-GrQc, spread mean	node	node+edge	node+edge+subgraph	gain from richness
r=1	0.9207	0.9204	0.9203	-0.0004
r=2	0.9416	0.9415	0.9414	-0.0002
r=3	0.9428	0.9427	0.9428	0.0000

This contradicts our own prior expectation that the depth-versus-richness trade-off would be interesting. On present evidence the surface collapses to one axis: *the neighbourhood you can see matters far more than how cleverly you describe it*. Reported as a negative result. Caveat: our subgraph tier is currently thin (triangles and ego-betweenness); adding graphlet orbit counts via ORCA is the fair test before concluding.

21. Finding 4 — the horizon is a property of the network

On the dense network, degree alone reaches $\tau = 0.85$ for spread and everything saturates at $r = 1$. On the sparse network, degree reaches 0.78 for spread but only 0.54 for betweenness, and saturation comes later.

The locality horizon is not a constant. It depends on the network's density and structure — which is the strongest argument that measuring it, rather than assuming it, is worthwhile.

22. Finding 5 — discriminability peaks near criticality

Degree-versus-spread rank correlation on email-Eu-core, as p sweeps through multiples of β_c :

multiple of β_c	0.5	1.0	1.5	2.0	3.0	5.0
$\tau(\text{degree, spread})$	0.916	0.860	0.880	0.906	0.938	0.960

The correlation dips to a **minimum near criticality**. That is exactly the regime where looking beyond degree pays most — and therefore the regime where our question is most interesting. It also justifies running at a small multiple of β_c rather than at an arbitrary p .

PART VIII — GLOSSARY

Every technical term and mathematical expression used in this document, in plain language.

23. Network structure

Term	Plain meaning
Node (vertex)	One entity in the network — a person, author, or account.
Edge (link, tie)	A connection between two nodes, such as a friendship.
Undirected	Edges have no direction: if A knows B, then B knows A.
Directed	Edges point one way, like a follow. We symmetrize these into undirected.
Degree , k	How many direct connections a node has.
Neighbour	A node directly connected to you.
Hop	One step along an edge. Two hops = a friend of a friend.
r-ball	Everything within r hops of a node.
Shell at distance r	Nodes at <i>exactly</i> r hops — the ring, not the filled circle.
Ego-network	A node plus its neighbours plus all edges among them.
Ego / alter	The ego is the node at the centre; alters are its neighbours.
Adjacency matrix , A	A table where entry (i,j) is 1 if i and j are connected, 0 otherwise.
Sparse matrix	A storage format that records only the non-zero entries; essential when most pairs are unconnected.
Path	A route from one node to another along edges.
Shortest path	The route using the fewest edges.
Connected component	A group of nodes all reachable from one another.
Largest connected component (LCC)	The biggest such group; we restrict analysis to it.
Self-loop	An edge from a node to itself. Removed.
Multi-edge	The same pair connected more than once. Collapsed to one.
Triangle	Three nodes all mutually connected.
Clustering coefficient , C	Of all pairs of your neighbours who <i>could</i> know each other, the fraction who do.
Density	Actual edges divided by possible edges in a group.
Assortativity	Whether nodes tend to connect to others of similar degree.
Structural hole	A gap between two of your contacts who do not know each other — a brokerage opportunity.
Effective size (Burt)	Your number of <i>non-redundant</i> contacts: neighbours who bring genuinely new reach.

Term	Plain meaning
Boundary porosity	How many edges lead out of your ego network — how "leaky" your local view is.
Growth ratio	How much bigger each successive shell is than the last; how fast your reach expands.
Hub	A node with unusually high degree.
k-core / coreness	The deepest densely-connected layer a node survives in when you repeatedly peel off low-degree nodes.
Graphlet	A small connected subgraph (3–5 nodes) — a structural motif.
Orbit	A distinct structural position within a graphlet.

24. Centrality measures

Term	Plain meaning	Local or global?
Degree centrality	How many connections you have.	Local
Betweenness centrality, $b(v)$	The fraction of all shortest paths that run through you. Measures brokerage.	Global
Closeness centrality	How close you are on average to everyone else.	Global
PageRank	You matter if people who matter link to you; defined recursively.	Global
Eigenvector centrality	Same recursive idea as PageRank, different formulation.	Global
Katz centrality	Importance from all paths reaching you, with longer paths counting less.	Global
Harmonic centrality	A closeness variant that copes with disconnected graphs.	Global
Ego-betweenness	Betweenness computed <i>only inside</i> your own ego network.	Local
H-index of a node	Apply the h-index idea to your neighbours' degrees instead of citation counts.	Local (per order)
Collective Influence, CI_ℓ	Your degree minus one, times the same quantity summed over the shell at distance ℓ .	Local

25. Spreading processes

Term	Plain meaning
Cascade	One run of something spreading through the network.
Seed	The node where the spread starts.
Independent Cascade (IC)	Each newly activated node gets one attempt to activate each neighbour, succeeding with probability p .

Term	Plain meaning
SIR	Susceptible–Infected–Recovered: an epidemic model where infected nodes eventually recover and stop spreading.
Linear Threshold (LT)	An alternative model where a node activates once <i>enough</i> of its neighbours have.
Transmission probability, p or β	The chance that one contact passes the infection along.
Reached set, $R(i)$	The set of nodes eventually activated when i is the seed.
Expected spread, $\sigma(i)$	The average size of $R(i)$ over many runs.
Epidemic threshold, β_c	The tipping point: below it outbreaks die out, above it they can engulf a large fraction of the network.
Criticality	Being at or near the tipping point, where behaviour is most variable and hardest to predict.
Percolation	Randomly keeping or deleting edges and asking what stays connected.
Bond percolation	The edge-deletion version; mathematically identical to IC with fixed p .
Live-edge graph	One sample of which edges successfully transmit.
Mean-field estimate	A simplified threshold formula that ignores network correlations; inaccurate on sparse graphs.
Ignition probability	The chance a cascade exceeds some large size instead of fizzling.

26. Linear algebra and spectra

Term	Plain meaning
Eigenvalue	A number describing how much a matrix stretches a particular direction.
Leading eigenvalue, λ_1	The largest one; it governs the long-run behaviour of repeated application.
Eigenvector	The direction associated with an eigenvalue.
Spectral radius	The magnitude of the leading eigenvalue.
Localisation	When the leading eigenvector concentrates on a few nodes (usually hubs), making estimates unreliable.
Non-backtracking walk	A walk that never immediately steps back where it came from.
Hashimoto / non-backtracking matrix, B	The matrix encoding those non-reversing steps, defined on directed edges.
Ihara-Bass identity	A result letting you get B 's leading eigenvalue from a smaller $2n \times 2n$ matrix.
Power iteration / Arnoldi	Numerical methods for finding the leading eigenvalue of a large sparse matrix.
Identity matrix, I	Ones on the diagonal, zeros elsewhere; the "do nothing" matrix.

Term	Plain meaning
Degree diagonal, D	A matrix with each node's degree on the diagonal.

27. Machine learning

Term	Plain meaning
Feature	One input number describing a node.
Target	The value we are trying to predict.
Regression	Predicting a number (rather than a category).
Random forest	A model that averages many decision trees; a strong, cheap general-purpose regressor.
Training / test set	Data the model learns from, versus data used to judge it.
k-fold cross-validation	Split into k groups; train on k−1 and predict the rest, rotating so every node is predicted once.
Out-of-fold prediction	A node's prediction made by a model that never saw it during training.
Overfitting	Memorising the training data instead of learning a general pattern.
Leakage	Accidentally letting the answer into the inputs; produces excellent, meaningless results.
Permutation importance	Shuffle one feature column and see how much performance drops; measures that feature's contribution.
Baseline	A simple rival method your approach must beat to be worth anything.
Heteroscedastic	Spread that changes with the size of the value — big means have big variance.
Residual	What is left over after subtracting the expected part.
Monte Carlo	Estimating a quantity by random simulation repeated many times.
Standard error, SE	How uncertain an average is, given how many samples produced it.

28. Evaluation metrics

Term	Plain meaning
Kendall's τ	Over all pairs, did we put them in the same order as the truth? +1 perfect, 0 random.
Concordant / discordant pair	A pair ranked the same way as truth / the opposite way.
Spearman correlation	Rank correlation; similar spirit to τ , computed differently.
Precision@k	Of the true top k, how many did our top k catch?
RMSE / MAE	Average size of numerical error; misleading on skewed targets, so kept secondary.
R²	Fraction of variance explained. Secondary here.

Term	Plain meaning
Coefficient of variation (CV)	Standard deviation divided by mean — relative variability, comparable across scales.
Skew	Asymmetry of a distribution; influence is heavily right-skewed.

29. Project-specific terms

Term	Plain meaning
Local feature	Computable from a bounded neighbourhood only — the only thing allowed as model input.
Global measure	Requires the whole graph; may be a target or a rival baseline, never an input.
Locality horizon, $r^*(\epsilon)$	The smallest radius whose performance reaches $(1-\epsilon)$ of the best we measured.
Locality budget	The general question of how much neighbourhood you can afford to look at.
Locality gap, $g_r(v)$	The fraction of a node's influence that comes from structure beyond r hops.
Depth vs richness	Looking further out (more hops) versus describing the same view more elaborately (more tiers).
Tier	Whether a feature describes a node, its edges, or small subgraphs it sits in.
Hop tag	The smallest radius at which a feature can be computed.
Depth = radius	An r -layer graph neural network sees exactly the r -ball, so "how far to look" and "how deep a network" are the same question.
Failure atlas	The profile of nodes local prediction gets most wrong, and what they have structurally in common.
Hidden influencer	A node whose true influence is high but which local features under-predict.
Archetype node	A node scoring very differently on two influence definitions — e.g. strong bridge, weak spreader.
Provenance record	The log of exactly what preprocessing was applied to a network.

30. Mathematical notation

Symbol	Meaning
n	Number of nodes
m	Number of edges
k_i	Degree of node i
$\langle k \rangle$	Mean degree across all nodes
$\langle k^2 \rangle$	Mean of squared degree; large when hubs are present
A	Adjacency matrix

Symbol	Meaning
B	Non-backtracking (Hashimoto) matrix
D	Degree diagonal matrix
I	Identity matrix
λ_1	Leading (largest) eigenvalue
σ_{st}	Number of shortest paths between s and t
$\sigma_{st}(v)$	How many of those pass through v
$b(v)$	Betweenness centrality of v
$\sigma(i)$	Expected spread when i is the seed
$\hat{\sigma}(i)$	Our Monte Carlo estimate of it
$R(i)$	The set of nodes reached from seed i
$ R(i) $	The size of that set
$E[\cdot]$	Expected value — the long-run average
p, β	Transmission probability
β_c	Critical threshold
μ	Recovery probability in SIR
M	Number of simulation runs
s	Sample standard deviation
SE	Standard error
r	Radius, in hops
$r^*(\epsilon)$	Locality horizon at tolerance ϵ
ϵ	Tolerance — how far below the ceiling we accept
$P(r)$	Best performance achievable using only the r -ball
τ	Kendall's tau
ρ	Fraction of edges deleted (robustness experiments)
$g_r(v)$	Locality gap of node v at radius r
$\Delta(v)$	Rank difference — used for both archetypes and failures
$Cl_\ell(i)$	Collective Influence of i at radius ℓ
$h^{(n)}(i)$	Order- n H-index of node i
$H(\cdot)$	The H-operator
Σ	Sum over the indicated range
$\in$	"is a member of"
$\subseteq$	"is a subset of"

Symbol	Meaning
$\cap$	Set intersection — elements in both
$\neq$	"not equal to"
$\geq, \leq$	"greater/less than or equal to"
$\rightarrow$	"maps to" or "step to"

PART IX — RISKS AND OPEN QUESTIONS

Risk	Mitigation
Two networks is a small corpus	Expand to the full pilot set before drawing firm conclusions
Single-graph CV is not true transfer	Train and test on separate networks as the corpus grows
Depth-beats-richness may reflect a thin subgraph tier	Add ORCA graphlet orbits and re-test before concluding
r^* is sensitive to ϵ	Report the whole curve and marginal gains, not a single number
Volatility result may be the mean in disguise	Variance-residual control implemented; run it and report both
Cascade estimates too noisy	Standard-error report built into every run; ratio currently 0.0036
Feature leakage	<code>assert_no_leakage</code> runs on every grid cell, not once

Open decisions:

- Which edge-probability convention to headline (weighted cascade is the most defensible default for social networks; uniform is what the percolation shortcut supports).
- Whether to add a threshold model (Linear Threshold) as a second dynamics arm.
- When to bring synthetic generation back in for the criticality sweep.
- Whether 5-node graphlets justify their cost over 4-node.

End of study document.